

Nolan Harris

Boulder, CO — (737)-529-1908 — nolandementharris@gmail.com — linkedin.com/in/nolanharris1 — GPA: 3.476

Education

University of Colorado at Boulder

B.S., Aerospace Engineering

May 2026

Boulder, CO

Relevant Coursework: Medicine in Space Environments; Orbital Mechanics; Aircraft Dynamics; Dynamics Controls Lab; Electronics; Thermodynamics and Heat Transfer; Structures; Aerodynamics; Foundations of Propulsion

Computer Science Minor

- C/C++, MATLAB, and Python programming with low-level Assembly processes
- Coursework: Algorithms; Data Structures; Computer Systems; Database Systems; Computational Methods

Skills

Programming: C, C++, Python, MATLAB, MySQL

Software: Adobe Photoshop/Lightroom, GitHub, SolidWorks, OpenVSP, EAGLE, LaTeX/Overleaf, MATLAB Simulink

Hardware: Rapid Prototyping, 3D Printing, PCB Manufacturing, Woodworking, Machine Shop, Soldering

Technical Experience

• Retractable Canard Passenger Aircraft:

- Using ArduPilot and Inav firmware on fixed wing prototype for flight control, telemetry, and video.
- Prototyping retractable canard on sub-scale technological demonstrator to analyze stability behavior in the transient state.
- Performed control analysis in MATLAB to determine control surface sizing and modal behavior.
- Present paper at AIAA student conference and present work to professors in design review presentations.

• EVA Medical Decision Aid (Mars Desert Research Station)

- Validated aid in an analog environment at the MDRS to evaluate performance in simulated conditions.
- Developed a risk decision matrix by organizing NASA medical standards into a weighted model to quantify likelihoods.
- Engineered an interface to reduce cognitive load, enabling medical triage and critical decision-making during EVAs.

• Arthritis Assisting Glove:

- Designed and manufactured custom PCB from EAGLE CAD
- Integrated flex sensors with an Arduino in a wearable glove for assisted finger movement

• Boost-glide project:

- Modeled an air propelled bottle rocket using predictive simulation in MATLAB.
- Validated against experimental data for existing gliders and aircraft.
- Iteratively fabricated design in machine shop and performed a boost-glide test.

Work Experience

Epic Productions

Lead Wedding Photographer

May 2022 – Present

Boulder, CO

- Manage wedding projects through planning and execution, ensuring client requirements and deadlines were met.
- Use Adobe and Microsoft tools for time management and project planning.

City of Austin

Head Lifeguard

May 2021 – Aug 2023

Austin, TX

- Maintained emergency procedures and risk assessment as well as managing scheduling.
- Instructed swim lessons for children and coached for youth swim team.

Austin Tri-Cyclist

Bike Mechanic

May 2019 - August 2022

Austin, TX

- Managed retail of triathlon gear including restocking and inventory logistics.
- Maintained, built, and sold bikes for triathlon athletes and mountain bike team.
- Performed maintenance on high, delicate carbon fiber frames to specification.

Leadership and Activities

AIAA

Officer

Nov 2024 – Present

Boulder, CO

- Organize campus networking events between students and Aerospace industry
- Represent CU at conventions and industry events

Westlake Video Crew

Crew Head

Aug 2019 – Dec 2021

Austin, TX

- Stage Manager for crew of 40+ members over 6 sold-out technical productions
- Performed safety briefings and the coordination of a full orchestra, a dozen sub crews, and first responders

Achievements

- **1st Place Freshman Projects Showcase:** Arthritis Assisting Glove
- **1st Place UIL Broadcasting 2018:** Westlake Video Crew